Final Report – Business Understanding

Idea Generation



CREATING MEANINGFUL EXPERIENCES

Applied Data Science & AI

Date

Breda University OF APPLIED SCIENCES

Index

Task 1.1: Market Research & Problem Definition	**2**
Market Research	**Error! Bookmark not defined.**
Problem Definition	**Error! Bookmark not defined.**
Task 1.2: Stakeholders	**3**
Task 1.3: DAPS Diagram	**6**
Task 1.4: Idea Generation	**8**
References	**9**

Task 1.1: Market Research & Problem Definition

A. Market Research

Conduct simple market research by answering the following questions. Use statistical data and justifications to support your answers:

- **What specific problems can my product or service solve for consumers?**

➢ Many businesses, including CRISP, where I work as a product controller, can benefit from an AI-powered food freshness detector. It can help us make better decisions, reduce waste, and save money.
➢ Food poisoning is a big problem. Every year, 48 million people in the U.S. get sick from bad food, leading to 128,000 hospital stays and 3,000 deaths (CDC, 2021). A food freshness detector can help stop this by warning people when food has gone bad.
➢ A lot of food is wasted. About 1.3 billion tons of food is thrown away every year—one-third of all food produced (FAO, 2020). This waste costs $940 billion a year and harms the environment (UNEP, 2021).
➢ Food freshness confusion leads to waste. Studies show that around 60% of consumers cannot correctly judge food freshness. As a result, they often throw away food that is still safe to eat or keep food that is already spoiled (WRAP, 2022). This lack of accurate freshness detection is a key driver of food waste and health risks.
➢ Restaurants and grocery stores lose money. In the U.S., food businesses lose $162 billion each year because of wasted food (ReFED, 2022). A freshness detector could help them manage food better and save money.
➢ Expiration dates are often wrong. Studies show 90% of people don't understand food labels, so they throw out food too early (Harvard Food Law and Policy Clinic, 2020). A freshness detector can tell if food is actually fresh instead of relying on printed dates.

- **What gaps or challenges exist in the current market or among competitors?**

➢ At CRISP as a product controller, it is hard for us to decide whether to throw away fruits or keep them. Sometimes, good fruits get thrown away because they look bad, and other times, spoiled fruits are kept because they look fine.
➢ Most food freshness checks are done by hand instead of using AI technology.
➢ There is no easy way for everyday people to check food freshness in real time.
➢ Food freshness sensors (like gas or chemical sensors) are too expensive and not widely used.
➢ Supermarkets and suppliers don't have the best methods for keeping track of food, leading to waste.
➢ Food safety rules are different in each country, making it hard to create a single global solution.

- **Are existing products or services meeting consumer expectations? If not, what are they lacking?**

- Some solutions, like smart labels and RFID sensors, help track food freshness but aren't always accurate or easy to use.
- AI technology for food analysis is mostly used in big food factories, not for everyday consumers.
- Most people still check food freshness by looking at it or smelling it, which isn't always reliable.
- High-tech freshness sensors exist, but they are expensive and not available for homes or small businesses.

B. Problem Definition

Based on the research conducted, create a problem definition. Justify it using conducted market research and statistical data.

Many people and businesses, including CRISP where I work as a product controller, find it hard to know if food is still fresh. This leads to food waste and health problems. Most people rely on looking, smelling, or checking expiration dates, but these methods are often inaccurate. According to the CDC (2021), millions of people get sick from spoiled food each year. Additionally, around 1.3 billion tons of food are wasted globally every year, causing environmental harm and financial loss (FAO, 2020). Expiration labels are often wrong, and most freshness checks depend on human judgment instead of technology. AI-based food analysis exists but is mostly used in large factories and not for everyday consumers. There is a need for an AI-powered image-based tool that helps people and businesses, like CRISP, check food freshness accurately, reducing waste, improving food safety, and saving money.

Task 1.2: Stakeholders

- **Create a list detailing each stakeholder's characteristics, encompassing aspects such as their power, influence, interest, expectations, needs, and preferences. Provide clear justifications when identifying stakeholders' power, influence, interests, expectations, needs, and preferences.**

1. **CRISP (Product Control Team)**

- Power: High - They decide how food is managed and handled in the company.
- Influence: High - Their choices affect food waste and business costs.
- Interest: High - They want to reduce waste, improve food safety, and manage food better.
- Expectations: They want an easy, accurate, and cost-effective tool to help with food decisions.
- Needs: A tool that helps them check food freshness in real-time.
- Preferences: They prefer a solution that is simple to use, saves time, and helps them make better decisions.

2. Consumers (General Public)

- Power: Moderate - They influence what products are popular through what they buy.
- Influence: Moderate - If consumers like the tool, it can spread and grow.
- Interest: High - They want to avoid eating spoiled food and save money by reducing waste.

- Expectations: They expect the tool to be easy to use, affordable, and accurate.
- Needs: A simple way to check if food is fresh.
- Preferences: They want a tool that's easy to use and helps them make better food choices.

3. Supermarkets and Retailers

- Power: High - They sell food and manage how it is stored and displayed.
- Influence: High - If they use the tool, it could help spread its use to other businesses.
- Interest: High - They want to reduce waste, improve stock management, and keep customers happy.
- Expectations: They want a cost-effective tool that helps with food management.
- Needs: A tool that helps them know when food is no longer fresh and reduces waste.
- Preferences: They prefer something that is easy to implement and improves their operations.

4. Government

- Power: High - They create and enforce rules about food safety.
- Influence: High - Their rules can affect how the tool is used and accepted in the market.
- Interest: High - They want to make sure food is safe to eat.
- Expectations: They expect the tool to meet food safety standards.
- Needs: A solution that helps food businesses stay compliant with safety rules.
- Preferences: They prefer technology that is easy to verify and complies with regulations.

5. Food Producers and Suppliers

- Power: Moderate - They provide food to supermarkets but don't control final sales.
- Influence: Moderate - Their choice to use the tool can affect how food is sold in stores.
- Interest: High - They want to make sure their products are fresh when sold.
- Expectations: They expect to be able to track the freshness of food from production to sale.
- Needs: A way to monitor the quality and freshness of food as it moves through the supply chain.
- Preferences: They want technology that is easy to use and integrates well into their systems.
-
- **Using the above information, prioritize and categorize stakeholders within the *Business-to-Consumer Power-Interest Matrix* (Figure 1). Explain the relationship between stakeholders and their respective positions within the matrix.**
- **Power-Interest Matrix**

SUBJECTS Low Power High Interest **1- Low Power, High Interest (Subjects):** • **Consumers:** They want to use the tool to check food freshness and avoid health issues but don't have the power to influence its use. • **Food Producers and Suppliers:** They care about food freshness but don't control how the tool is used.	**PLAYERS** High Power High Interest **2- High Power, High Interest (Players):** • **CRISP (Product Control Team):** They make decisions about food and waste and need the tool to make better choices. • **Supermarkets and Retailers**: They want to reduce waste and manage stock. They can influence the product's success.
CROWD Low Power Low Interest **3- Low Power, Low Interest (Crowd):** • **Tech Developers:** They create the tool but don't directly influence its success or the market.	**CONTEXT SETTERS** High Power Low interest **4- High Power, Low Interest (Context Setters):** • **Food Safety Regulators (Government Bodies):** They set rules for food safety but may not care much about the tool. • **Investors:** They provide funding and care more about business growth than the specifics of the tool.

- **Identify the main stakeholder and explain their needs, priorities, and expectations, as well as the reason behind their need to solve the problem. Explain why it will be beneficial for the company to address this problem, focusing on this "main stakeholder". Use statistical data and justifications to support your answers.**

Main Stakeholder: CRISP (Product Control Team)

- **Needs:** CRISP needs a way to make better decisions about food freshness to reduce waste and ensure food safety.
- **Priorities:** Their top priorities are reducing food waste, saving money, and improving stock management.
- **Expectations:** CRISP expects an easy-to-use AI tool to help them quickly check food freshness and make better decisions.
- **Reason for Solving the Problem:**
- CRISP struggles to decide which food is fresh and which is not, leading to wasted food and lost money. They need a tool to help them check food quality in real time.
- **Why It's Beneficial for CRISP to Address This Problem:**
- **Save Money:** Reducing food waste would save a lot of money. The FAO (2020) says food waste costs $1 trillion globally each year.
- **Food Safety:** The CDC (2021) reports that many people get sick from spoiled food. CRISP can reduce this risk by using better tools to check freshness.
- **Reputation:** Solving this problem can help CRISP look good in the market and improve trust with customers.
- **Supporting Stats:**
- **Food Waste:** 1.3 billion tons of food are wasted worldwide every year (FAO, 2020).
- **Foodborne Illness:** Millions of people get sick from spoiled food each year (CDC, 2021)

Task 1.3: DAPS Diagram

Link to Task1.3DAP Diagram's On GitHub

Business value

Reduce Food Waste & Improve Food Safety

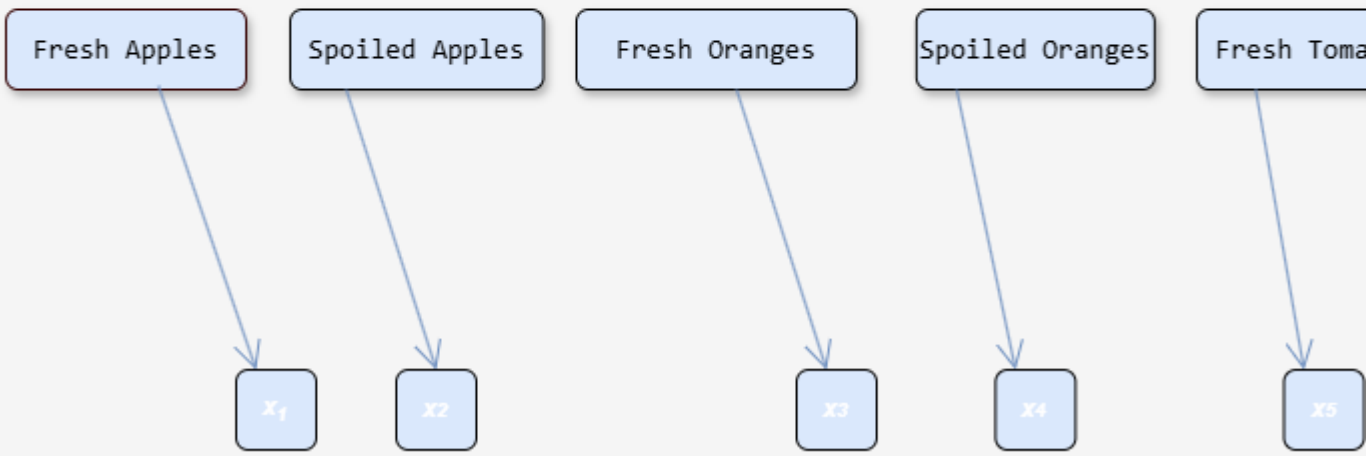
- **Improve Food Handling & Decision Making**
- **Improve Consumer Awaren**
- **Accuracy of AI Model**
- **Accuracy**

Image:



- **What the AI Does**:
  - Analyzes food images to check freshness.
  - Predicts if food is **fresh or spoiled**.
  - Recommends actions: **Keep or Discard** food.
- **Who Uses It?**
  - CRISP product control team
  - Supermarkets and retailers
  - Consumers
- **When Are Decisions Made?**
  - Before throwing away food.
  - Before selling or serving food.
- **How Does It Help?**
  - **Reduces waste** by preventing good food from being discarded.
  - **Improves safety** by detecting spoiled food early.
  - **Saves time and money** by making quick, accurate decisions.

Data-analytic problem



1 How will the project contribute to the organization's goals?
The project helps reduce food waste and improve food safety by using AI to check if food is fresh or spoiled. This supports the organization's goal of saving money, protecting customers' health, and making better decisions about food handling.
2 What is the direct benefit of applying the decision framework?
It helps the organization make faster and smarter decisions about food freshness. This reduces waste, saves money, and keeps food safe for customers.
3 For what decisions or actions will the model be used?
The model will help decide if food is fresh or spoiled and whether to keep or throw it away.

- **What agent takes decisions?**
- The decision is made by the product control team at CRISP, supermarket staff, or even regular users.
- **When are decisions taken?**
- Decisions are taken before selling, serving, or throwing away food.
- **What is the unit of analysis?**
- Each image of a food item is one unit of analysis.
- **Intervention in x's?**

The intervention is applying AI to analyze image features (like color, texture) to predict freshness.
4 What will the model predict?
The model will predict if a food item is fresh or spoiled.
5 What is the type of relationship?
The relationship is correlational, as the model learns patterns between image features and the freshness label.
6 What are potential predictors, features?
The features (x's) include image-based cues like color, texture, shape, and surface appearance of the food.

Task 1.4: Idea Generation

Define the main idea for your application and align it with your market research, business value, and consumer needs.

Problem Definition & Link to the Final Product

Many businesses and consumers struggle to determine if food is fresh or spoiled. Relying on visual checks, smell, or expiration dates often leads to throwing away good food or consuming spoiled food, creating both waste and health risks. According to WRAP (2022), 60% of people misjudge freshness, and ReFED (2022) reports that U.S. food businesses lose $162 billion annually due to waste.

To address this, the final product is an AI-powered image-based food freshness detector. It uses deep learning to classify food items (such as apples, oranges, and tomatoes) as either *fresh* or *spoiled* based on uploaded images.

How the Product Addresses Stakeholder Needs

The main stakeholder, CRISP's product control team, faces challenges in making fast, consistent, and accurate food quality decisions. Currently, they manually inspect food items, which is:

- Time-consuming
- Inconsistent
- Prone to human error

The AI tool helps CRISP by:

- Allowing users to upload images of food items for real-time classification (e.g., "Fresh Apple", "Spoiled Tomato")
- Replacing guesswork with AI-driven decisions (keep or discard)
- Reducing waste from throwing away food that's still good
- Preventing health risks by catching spoiled items earlier
- Saving labor time through automation

Business Value of the Application

This AI solution provides several benefits for businesses and the food industry:

Reduces Food Waste – Prevents throwing away good food, saving money and resources.
Improves Food Safety – Detects spoiled food early, reducing health risks for consumers.
Increases Efficiency – Automates food quality checks, eliminating the need for manual inspections.
Enhances Brand Reputation – Businesses using AI for food quality can promote sustainability and food safety, attracting more eco-conscious customers

References

- Centers for Disease Control and Prevention (CDC). (2021). *Burden of foodborne illness: Findings*. https://www.cdc.gov/foodborneburden/index.html
- CRISP. (n.d.). *About us*. Retrieved from https://www.crisp.nl/en
- Food and Agriculture Organization of the United Nations (FAO). (2020). *The state of food and agriculture 2020: Overcoming water challenges in agriculture*. http://www.fao.org/publications
- Harvard Law School Food Law and Policy Clinic. (2020). *The Dating Game: How Confusing Food Date Labels Lead to Food Waste in America*. https://chlpi.org
- OpenAI. (2024). *About OpenAI*. https://openai.com/about
- ReFED. (2022). *Insights Engine: The Food Waste Monitor*. https://refed.org
- United Nations Environment Programme (UNEP). (2021). *Food Waste Index Report 2021*. https://www.unep.org/resources/report/unep-food-waste-index-report-2021
- Waste and Resources Action Programme (WRAP). (2022). *Citizen Food Waste Tracker Survey Results*. https://wrap.org.uk

- World Health Organization (WHO). (2021). *Food Safety*. https://www.who.int/news-room/fact-sheets/detail/food-safety



Games



Leisure & Events



Tourism



Media



Data Science & AI



Hotel



Logistics



Built Environment



Facility

Mgr. Hopmansstraat 2
4817 JS Breda

P.O. Box 3917
4800 DX Breda
The Netherlands

PHONE
+31 76 533 22 03
E-MAIL
communications@buas.nl
WEBSITE
www.BUas.nl

CREATING MEANINGFUL EXPERIENCES

Breda University
OF APPLIED SCIENCES